

DYNAMIC DECISION ALGORITHM

*A Recursive Bayesian Framework for
Adaptive Decision Optimization*

COMPLETE MATHEMATICAL PROOFS

Mathematical Framework

DDA v7.0 CORE FORMULATION
Classic EMA (COUPLED): $F_n = P_0 \cdot F_{n-1} + (1-P_0) \cdot I_n$ → Noise rejection and lag are COUPLED (change one = change both)
DDA v7.0 (DECOUPLED): Step 1: $\Delta_n = I_n - I_{n-1}$ (Raw derivative) Step 2: $\tilde{\Delta}_n = \alpha \cdot \Delta_n + (1-\alpha) \cdot \tilde{\Delta}_{n-1}$ (Filter derivative) Step 3: $L_n = I_n + \gamma \cdot \tilde{\Delta}_n$ (Pre-cognitive boost) Step 4: $F_n = P_0 \cdot k \cdot F_{n-1} + m \cdot L_n$ (Bayesian update) → Noise (α) and Lag (γ) are INDEPENDENT!
Parameters: $P_0=0.70$ $\gamma=0.60$ $\alpha=0.10$ $k \in [0.9, 1.1]$ $m=1-P_0 \cdot k$